

Machine Learning - Assignment

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Objectives, First leg

Let's look into the assumptions and behavior of different machine learning methods. The **general objective** of this assignment is to dive a little bit deeper into how different ML methods behave on a dataset.

Tasks

1. Produce the following artificial datasets in balanced (50/50) and unbalanced (90/10) class distributions.
 - Headlights: 2 gaussian blobs same covariance with some overlap, 1 class each.
 - Shell: 1 circular gaussian blob (class A) and one somewhat rectangular thin strip of points with the width of the blob (class B) slightly overlapping the edge of the blob. Balanced classes.
 - DD: 1 half circular blob, D shaped (class A) and another similar one to the right almost touching but with almost no or minimal overlapping (class B)
 - Xor: Four alternating squares: a 2 by 2 chess board alternating classes A and B
 - Chess4: Like the one above but 4 by 4, still 2 alternating classes
 - Roseta2: 1 round central blob (class A), 1 ring around the central blob, like a blob without the central part, where class A is (class B)
 - Roseta3: Same as above, but 3 concentric rings with 2 alternating classes
2. Consider the methods: 1NN, 5NN, 10NN, logistic regression, LDA and QDA.
 - a. For each of the suggested datasets rank the **listed methods** by accuracy on an independent test set.
 - b. Explain the results taking into account the assumptions of the methods.
 - c. Suggest 1 dataset that produces a clearly different ranking. Explain the results.

Second leg, to be defined

Processing...

Submit

- A notebook with answers to these questions. The notebook is ipynb by default. Any other format must be easily readable. Each answer must be clearly identified and respect the order of the questions. Please take care with the following:
 - Steps taken must be succinctly described.
 - Results must be summarised as much as possible.
 - Well formatted and clear references to any paper, slides, book or site consulted for this assignment.
- You can also submit an html version of the notebook.
- Each group presents the work in-class by the end of the deadline. All elements of the group should participate in the presentation.

Suggested structure

- Authors clearly identified at the top of the notebook
- Answers
- References
- Appendixes (optional - non-essential add ons, longer and more detailed experiments, etc.)

Important

- Important notes (I apologize for the caps but these are recurrent issues):
 - ONLY MOODLE CAN BE USED FOR SUBMISSION. No email submission.
- Guidelines:
 - The answers should be clearly written.
 - The objectives for each experiment and plot you show should be clear.
 - It is not necessary to describe the methods but you should explain your steps when doing experiments and making choices.

Evaluation

- Components
 - Report (per group)
 - * Writing 30%
 - * Technical 70%
 - Presentation (per student)
 - * Clarity, confidence, communication, timely delivery.50%
 - * Technical 50%
 - Final grade is Report * Presentation

Groups

Assignments are submitted by groups of 2 to 3 students. Different elements may have different grades based on the contribution distribution and interactions about the assignment. This distribution can be declared by the members of the group or can be inferred by the lecturers on the basis of individual interaction. Other group sizes will not be considered.

It is advisable that the students from the same group share general tasks but may have complementary more specific tasks. Group work is important for learning from other people, but each individual must acquire independent skills.

Submissions

Formal final deadline is **June 1st 2026**, to be submitted in moodle, and only in moodle. Submissions after that date will be multiplied by a monotonously decreasing factor that starts in 1 with a negative slope of .15 values per day until a minimum of 50% of the original value.

- Checkpoints:
 - In the class of the **28th April** there will be a checkpoint for this leg and some more. Each group should present on desk an update with the status of the project. You will have around 5 minutes for this presentation.
 - Final presentations of the project on **June 2nd** morning and afternoon. All members of the group should be present. No-shows get a zero.

Use of resources including Generative AI (pedagogical advise)

You, students, are learning how to do data science and the main aim of this exercise is that you gain skills. Skills are gained with different depth and robustness depending on how we use the learning resources. Effort must always be there. Beware of too easy routes - they usually teach us nothing. No pain, no gain.

Your **resources** are class materials, books, youtube videos, materials on the web, etc. (CM), programming documentation, such as the manuals of packages you use, search engines, e.g. google (SE) and generative AI, e.g. claude, mistral, chatgpt or gemini (GAI).

Our **advise** is that you approach your problems by using these resources by this order and with different types of usage. When thinking of concepts, use CM, when you need to know how to use a function or what to use in a particular setting, use PD. SE are very important for more directed searches, looking for code snippets, looking for more materials, finding fruitful discussions. Finally GAI gives us more value when used for concept clarification or to find the starting point for a search or quest. The more you know when you use GAI, the more you take from it. If you take nothing, you learn nothing. Many other usages of GAI are productive but they should always have you leading, not the other way around.

An **example of good usage of GAI** is “Which hyperparameters are important for using BPR and why?”. A **bad usage** is “produce the code for applying BPR for this problem”. You can gain skills with the first type of usage whereas with the second you rely entirely on the skills of generative AI. That implies that any future employee will rather buy a license of Claude or chatGPT than hire you (because it is cheaper). Or they will hire someone with more skills that can validate and better exploit GAI. That is because when you gain skills you are ready to gain more skills and you are a valuable asset that can check what gen AI is producing. Having skills makes you think longer, deeper and depend less on externalities. Deep skills make you more robust and more resilient. You will be valuable even without GenAI. If your skills are easy to get then anyone cheaper than you can do it.

Ethical principles

When submitting, students commit themselves to follow strong ethical principles. All the work must be done by the elements of the group alone. All members of the group will be involved with the whole of the work. All the materials used and consulted must be credited in the work.

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